

CSA5103 - CRYPTOGRAPHY AND NETWORK SECURITY LIST OF PROGRAMS

EXPERIMENT 4

Write a C program for polyalphabetic substitution cipher uses a separate monoalphabetic substitution cipher for each successive letter of plaintext, depending on a key.

Aim: To write a Python program for the Polyalphabetic Substitution Cipher.

Algorithm:

Step 1: Read the plaintext and key from the user.

Step 2: Convert both the plaintext and key to uppercase.

Step 3: Repeat the key until it matches the length of the plaintext.

Step 4: Encrypt each plaintext letter using the corresponding key letter.

Step 5: Generate the ciphertext.

Step 6: Display the encrypted message.

Step 7: End.

Program:

```
text = input("Enter the plaintext: ").upper()
key = input("Enter the key: ").upper()
result = ""
for i in range(len(text)):
    p = ord(text[i]) - 65
    k = ord(key[i % len(key)]) - 65
    c = (p + k) % 26
    result += chr(c + 65)
print("Encrypted Text:", result)
```

Output:

```
Enter the plaintext: HELLO
Enter the key: KEY
Encrypted Text: RIJVS
```

Result:

The Python program for the Polyalphabetic Substitution Cipher was executed successfully, and the plaintext was encrypted correctly using the given key.